Higher K property: A is an n-type if and only if its (n+1)-fold loop space is contractible

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:ntype-nloop' (source lines 423-426, Theorem 7.2.9); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own remark that the generalization of the K property is obtained by induction and slightly clever whiskering, which is precisely where the difficulty of the proof sits.

Editorial note. The retained counter values (chapter 7, section 2, theorem offset 8) make the retained theorem print as Theorem 7.2.9, matching the original book numbering.

Editorial note. The five external references are navigation pointers to results used in the two base cases and in the construction of g; the counted proof is retained in full.

Editorial scope. One original result of the HoTT Book, counted once: the higher K property in terms of iterated loop spaces (Theorem 7.2.9).

Allowed standard prerequisites (editorial).

- Iterated loop spaces. Omega^0(A,a) := (A,a) as a pointed type and Omega^{n+1}(A,a) := Omega^n(Omega(A,a), refl a).. The proof manipulates Omega^{n+2}(A,a) by this definition at source lines 432-434.
- Truncation of loop spaces. A is an (n+1)-type if and only if Omega(A,a) is an n-type for all a.. It is the reformulation used at source line 431.
- Pointed types and equivalences of pointed types with a basepoint condition. An equivalence of pointed types must carry the basepoint to the basepoint.. The proof constructs g : Omega(Omega(A,a),p) ~ Omega(Omega(A,a),refl a) and checks g(refl p) = refl(refl a) at source lines 435-444.

Source reference. Original exercise 3.5: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1386–1386. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original theorem 7.2.1: sources/hott-hlevels/pinned-original-source.tex, source lines 247–247. Original result cited by the retained text but

located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original theorem 7.2.7: sources/hott-hlevels/pinned-original-source.tex, source lines 394–394. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original reference 2.4.8: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1197–1197. Original example (Example 2.4.8 of the book) cited by label in the retained proof; it is declared with the generic pointer kind because the item is an example rather than a lemma or theorem. It is a navigation pointer only: the counted proof is retained line for line and nothing is replaced by it.

Source reference. Original theorem 2.11.1: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1854–1854. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

By induction and some slightly clever whiskering, we can obtain a generalization of the K property to $n > 0$.

Theorem 7.2.9. *For every $n \geq -1$, a type A is an n-type if and only if $\Omega^{n+1}(A, a)$ is contractible for all $a : A$.*

Proof. Recalling that $\Omega^0(A, a) = (A, a)$, the case $n = -1$ is original exercise 3.5. The case $n = 0$ is original theorem 7.2.1. Now we use induction; suppose the statement holds for $n : \mathbb{N}$. By original theorem 7.2.7, A is an $(n+1)$-type iff $\Omega(A, a)$ is an n-type for all $a : A$. By the inductive hypothesis, the latter is equivalent to saying that $\Omega^{n+1}(\Omega(A, a), p)$ is contractible for all $p : \Omega(A, a)$.

Since $\Omega^{n+2}(A, a) :\equiv \Omega^{n+1}(\Omega(A, a), \mathsf{refl}_a)$, and $\Omega^{n+1} = \Omega^n \circ \Omega$, it will suffice to show that $\Omega(\Omega(A, a), p)$ is equal to $\Omega(\Omega(A, a), \mathsf{refl}_a)$, in the type $\mathcal{U}_\bullet$ of pointed types. For this, it suffices to give an equivalence

$$g : \Omega(\Omega(A, a), p) \simeq \Omega(\Omega(A, a), \mathsf{refl}_a)$$

which carries the basepoint refl_p to the basepoint $\mathsf{refl}_{\mathsf{refl}_a}$. For $q : p = p$, define $g(q) : \mathsf{refl}_a = \mathsf{refl}_a$ to be the following composite:

$$\mathsf{refl}_a = p \cdot p^{-1} \stackrel{q}{=} p \cdot p^{-1} = \mathsf{refl}_a,$$

where the path labeled "q" is actually $\mathsf{ap}_{\lambda r.\, r \cdot p^{-1}}(q)$. Then g is an equivalence because it is a composite of equivalences

$$(p = p) \xrightarrow{\mathsf{ap}_{\lambda r.\, r \cdot p^{-1}}} (p \cdot p^{-1} = p \cdot p^{-1}) \xrightarrow{i \cdot - \cdot i^{-1}} (\mathsf{refl}_a = \mathsf{refl}_a).$$

using original reference 2.4.8 and original theorem 2.11.1, where $i : \mathsf{refl}_a = p \cdot p^{-1}$ is the canonical equality. And it is evident that $g(\mathsf{refl}_p) = \mathsf{refl}_{\mathsf{refl}_a}$. □